

Table of Contents

Reporting of Analysis	2
Introduction	2
Business Problem:	2
Stakeholders	3
Data Source	4
Storyline and Visualization:	6
Objective of Analysis	5
Analysis	6
Data Model	6
Dashboard:	7
Sales Performance Dashboard.....	7
Seller Performance Dashboard.....	8
Delivery and Logistic Analytics	9
Customer Behavior and Review analysis Dashboard.....	10
Insights:	12
Recommendations	13

Reporting of Analysis:

The Brazilian E-Commerce Public Dataset is a comprehensive collection of retail data encompassing 100,000 orders from Olist, placed between October 2016 and September 2018 across diverse states. This dynamic dataset evolves over time as new information is progressively added. It provides a wealth of metrics such as pricing, order numbers and statuses, payment methods, freight costs, and customer reviews, offering a well-rounded understanding of e-commerce patterns and trends.

Introduction:

Businesses have consistently strived to retain their customer base's engagement and satisfaction with the services they offer. To stay competitive in today's market, it is crucial for them to incorporate the latest technological advancements into their service delivery mechanisms.

Over a decade ago, when the Internet was still an emerging technology, numerous industries sought to leverage its potential as a communication conduit between various enterprises. This was a formative step in harnessing the power of digital transformation.

In the current decade, industries have evolved to provide services that are intricately tailored to meet the unique requirements of each customer. This level of personalization and precision necessitates the use of artificial intelligence, which is now integral to offering such tailored services.

Business Problem:

Olist, an online store with its corporate headquarters in Sao Paulo, Brazil, serves as a central hub for numerous small businesses and customers seeking to purchase their products. The company has shared a dataset on Kaggle containing details of 100,000 orders placed across various marketplaces between 2016 and 2018.

Customer reviews on e-commerce websites significantly influence our purchasing decisions. As such, Olist can utilize this feedback to streamline their product offerings, potentially eliminating products that consistently receive poor reviews. Conversely, they could choose to promote products that have been well-received by consumers.

A Power BI-generated dashboard provides a comprehensive view of overall customer satisfaction with purchased products. This platform employs both descriptive and predictive analysis to give a detailed account of past consumer behaviors and forecast future trends.

Stakeholders:

Olist's organizational structure is clearly defined and hierarchical, ensuring efficient administration and smooth operations. Clear communication channels are made possible by the organization's hierarchical structure, ensuring that every decision made at the top is effectively trickled down.

The CEO, who acts as the main decision-maker, is at the top of this structure. The CEO's choices, from evaluating and approving budgets to assuring the effectiveness of subordinate teams, are essential to the direction of the business. As a shareholder, the CEO, who is frequently the chairman or owner, has a large financial stake in the business.

In the hierarchy, the Marketing Team is in second place. The promotion of Olist's goods and services is greatly aided by this team, which uses cutting-edge technology to remain competitive. As stakeholders, the marketing division makes sure the business is visible and appealing to potential customers.

The Finance Team comes after the Marketing Team. The management of Olist's accounts and maintenance of the company's profitability depend on the work of this team. They manage cash flows, debts, loans, depreciation, income, taxes, and other financial matters while staying within set budgetary parameters. Additionally, this team offers monetary options like conditional coupons to boost sales or control the marketing budget necessary for new projects.

The Information Technology (IT) Team is the next group, which is essential to any e-commerce firm. Furthermore, the Olist IT infrastructure is managed by this team, which also provides assistance to other teams inside the company. They are essential to maintaining e-commerce operations efficiently and without hiccups. The organization's E-commerce Team, another essential part, manages the online sales procedure. They engage with consumers, make sure transactions go well, and help to increase customer loyalty and retention.

The CEO, Marketing, Finance, IT, and E-commerce departments, however, are the main parties participating in and impacted by this initiative.

It is easy to pinpoint the stakeholders who would be most impacted and invested in the project's outcomes by comprehending this organizational structure. For the project to be a success and for Olist to continue to develop and prosper, it is essential to address their particular needs and concerns.

Data Source:

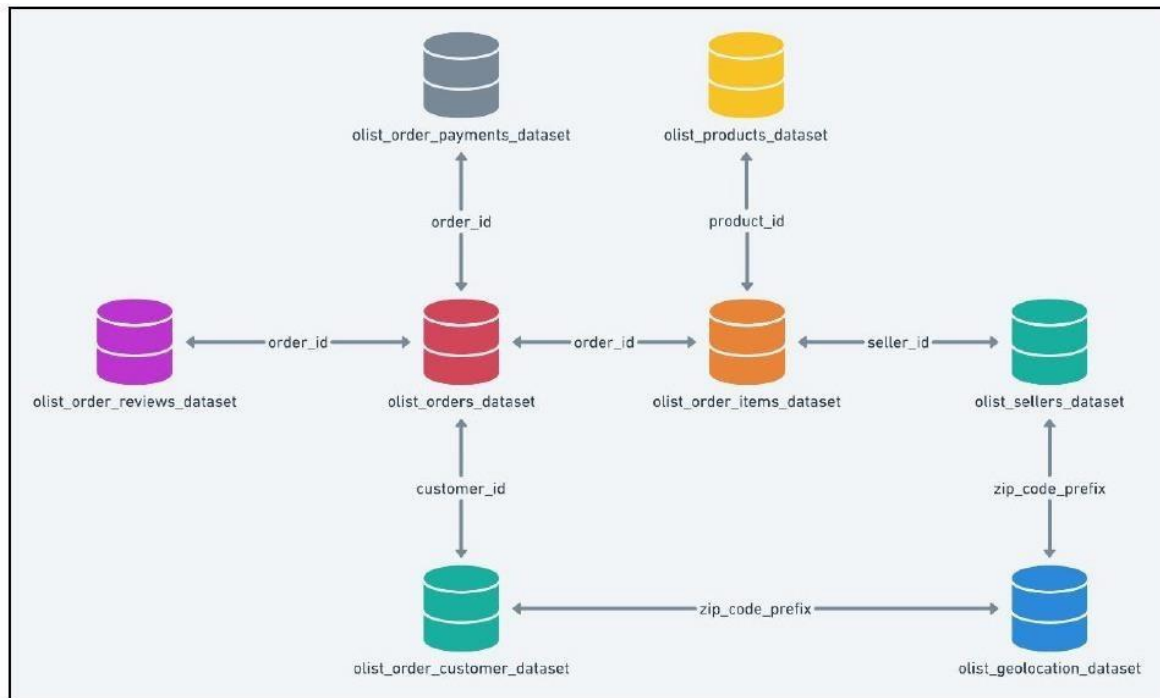
The plan is to leverage the dataset generously provided by Olist, the largest department store in Brazilian marketplaces. Olist functions as a crucial link, connecting small businesses from all over Brazil to various distribution channels, thereby streamlining administrative tasks and consolidating legal obligations. This setup allows merchants to sell their products through the Olist Store. Following a sale, Olist's logistics partners are responsible for the direct delivery of the items to the purchasers.

The data encompasses over 100,000 unique customer and order identifiers. It is comprehensively structured to include detailed information on orders, customers, products, sellers, payments, geolocations, and reviews, offering a holistic view of the entire commercial process.

Figure 2 Description of tables

Table	Description
olist_orders_dataset	It is linked to 4 table and connects all the information pertaining to an order.
olist_order_items_dataset	It includes information on a purchased item, including the price, shipping date, and other facts.
olist_order_reviews_dataset	It includes information about any reviews that the customer may have written about a certain item that he had purchased. It includes information about a product, including its ID, category name, and measurements.
olist_products_dataset	It includes information about a product, including its ID, category name, and measurements.
olist_order_payments_dataset	This table's data relates to the payment-related information connected to a certain order. Describes this company's customer base in detail.
olist_customers_dataset	Describes this company's customer base in detail.
olist_sellers_dataset	All of the sellers who have registered with this company are listed in this table along with their connected information.
olist_geolocation_dataset	Both the geographical location of the buyers and the sellers is included.

Figure 1 Relationship Diagram (Entity Relationship)



Storyline and Visualization:

The line manager has requested the use of business intelligence tools to thoroughly analyze the provided datasets and present the findings and recommendations in a comprehensive report format. This dataset, rich with information on 100,000 orders placed across various Brazilian marketplaces, facilitates a multifaceted perspective of each transaction. Such insights include customer location, product attributes, order status, pricing, payment processes, and freight performance. Moreover, the dataset incorporates a geolocation component that associates Brazilian zip codes with corresponding latitude and longitude coordinates.

Once an order is placed through the Olist Store, the relevant vendor is notified to complete the order. After the product has been delivered or the projected delivery date has passed, the customer is invited to fill out a satisfaction survey via email. This process allows them to rate their shopping experience and offer constructive feedback.

For this analysis, Microsoft Power BI has been chosen as the tool for data visualization. This platform stands out due to its interactive dashboards and compatibility with a diverse array of data sources.

Objective of Analysis:

The main objective of the analysis would be able to answer below questions:

- a) What is the company's total number of customers, orders, and orders per customer?
- b) Which states have the most customers?
- c) How many orders were there each month?
- d) What are the top five product categories and the lowest five?
- e) Display the consumer demographics, sales trends, order categories, and order changes over time for the company.

Analysis:

Data Model:

The dataset, sourced from Kaggle, has been imported into Power BI. As it currently contains only tables and corresponding keys for each file of the dataset, pre-processing is required to ensure precise analytics. We commence by constructing a data model, the structure of which can be viewed in the accompanying Figure 3.

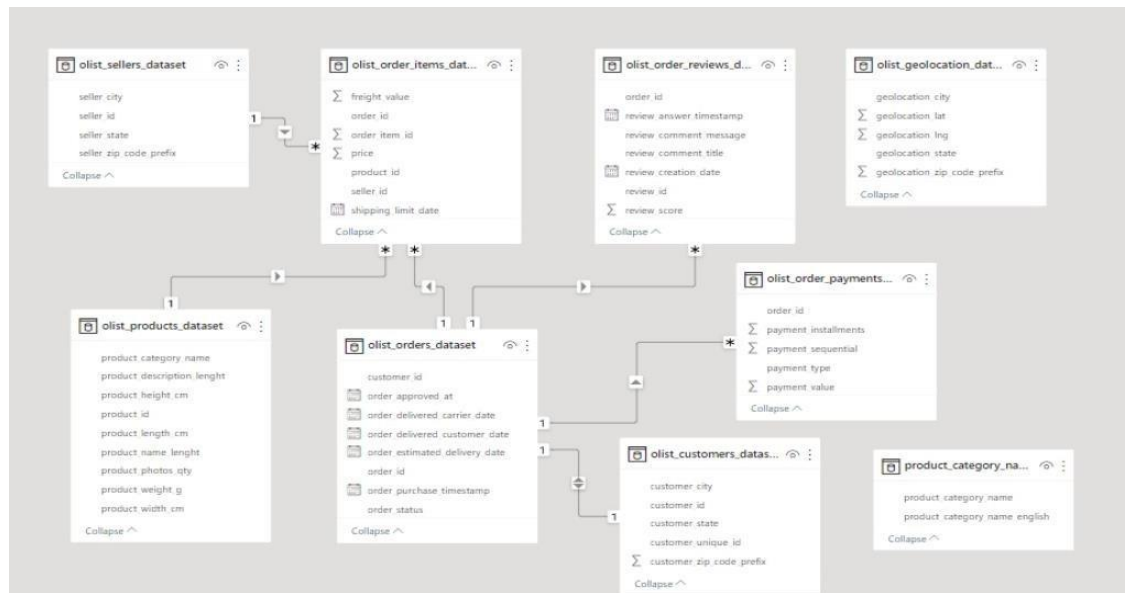
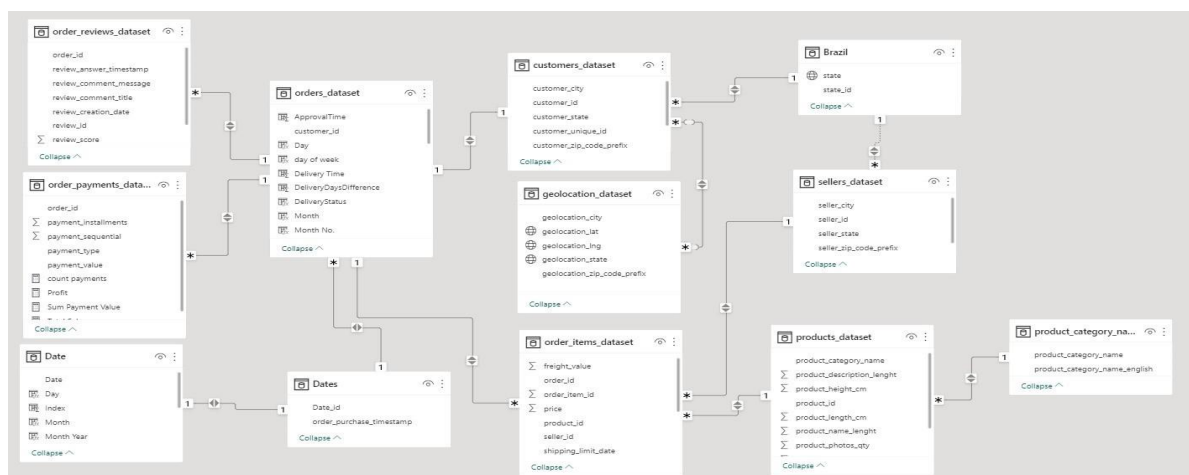


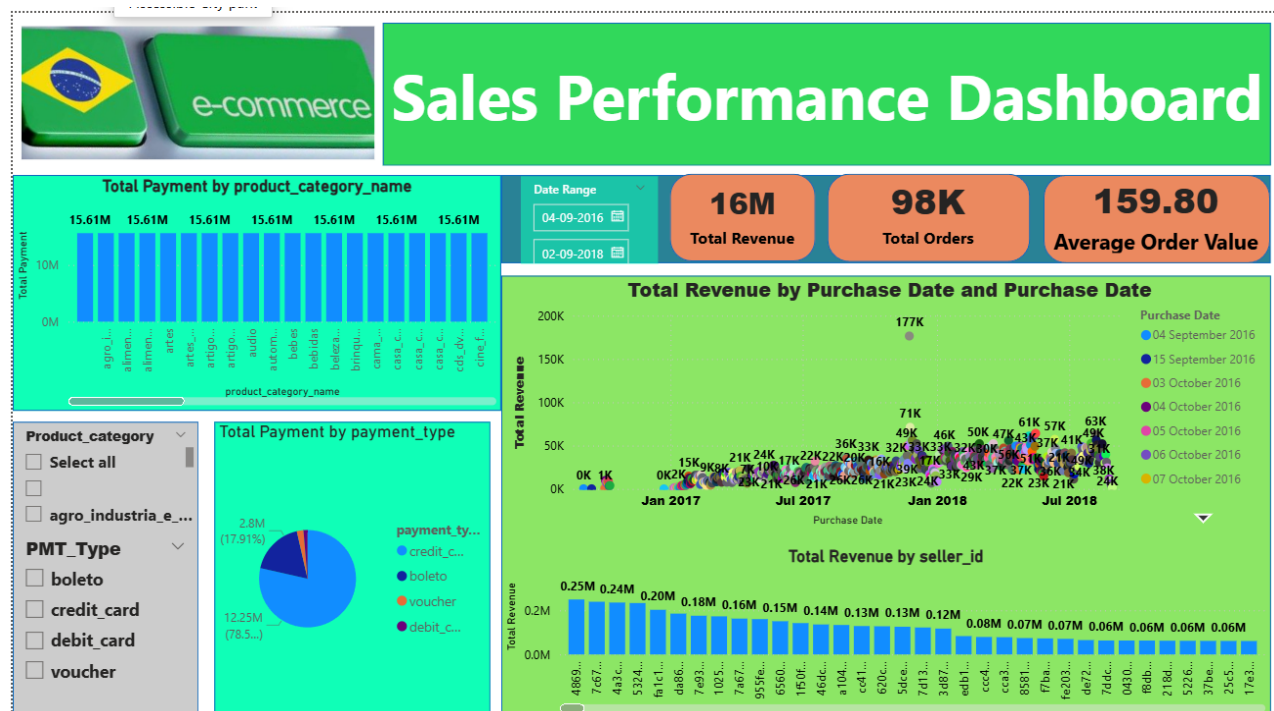
Figure 3 Initial schema

Consequently, depending on the interactions between the tables, connections or linkages between them are essential. The 'olist_geolocation' and 'product_category' tables cannot be incorporated as they don't have any apparent relation to the existing data model. Temporal information is also not explicitly stated as efficient time management is crucial when applying Power BI to the remaining tables. Furthermore, the variables are categorized based on the nature of the data they contain, whether it be textual or numerical. It's evident that the dataset encompasses information regarding transportation logistics, customers, vendors, and a range of products sold, each with a unique identifier. The final schema of the refined data model can be viewed in the following Figure 4.



Dashboard:

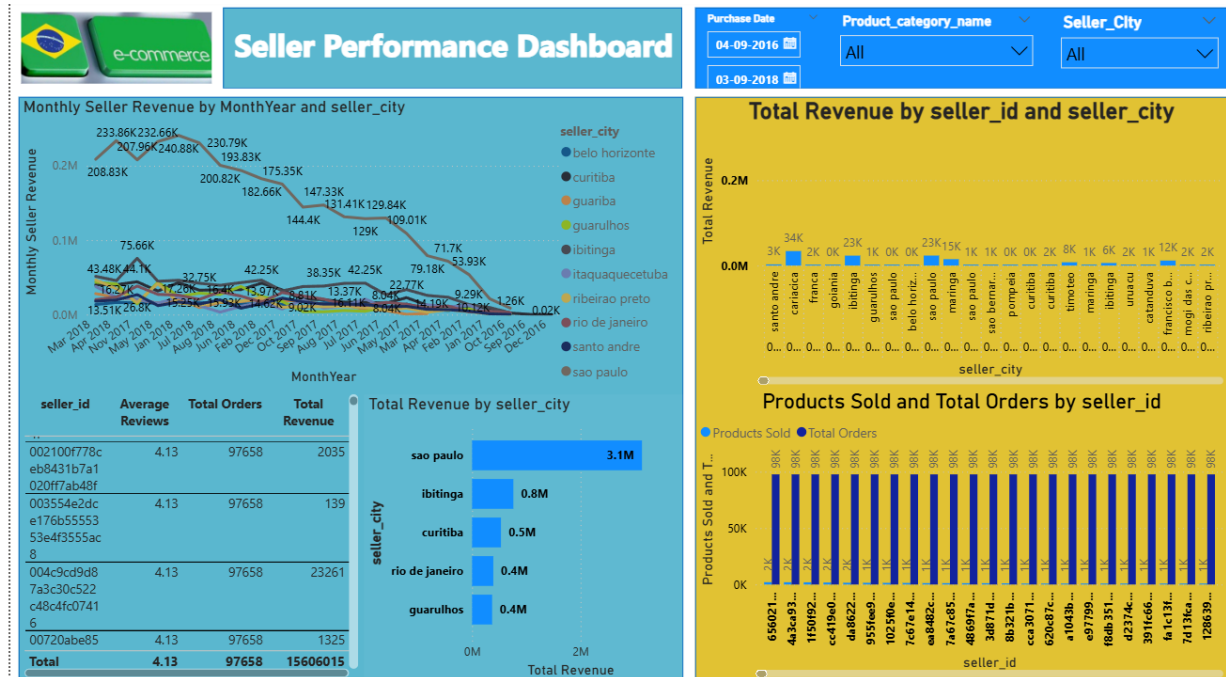
Sales Performance Dashboard



This dashboard is designed to provide a comprehensive view of the company's performance over the years by highlighting key metrics such as total revenues, total orders, Average Order Value, Total Payment with Product Category Name, and Payment Type. As these aspects form the core of any business operation, the focus lies on revenue generation and Total Orders. From 2016 to 2018,

The company has demonstrated significant growth, reflected in the data with an average of 16M in total sales and an order quantity of 98K. This sustained expansion signifies the company's successful strategies and effective operations.

Seller Performance Dashboard



Monthly Seller Revenue by Month and Seller City

This line chart reveals temporal sales trends segmented by seller cities. **Sao Paulo** consistently leads in monthly revenue, followed by cities like **ibitinga** and **Curitiba**. A sharp increase is observed during key months, indicating seasonal buying behavior.

Average Review Score Table

The table showcases performance at the seller level, including metrics like **Average Reviews**, **Total Orders**, and **Total Revenue**. Uniform review scores (e.g., 4.13) suggest consistent customer satisfaction across sellers.

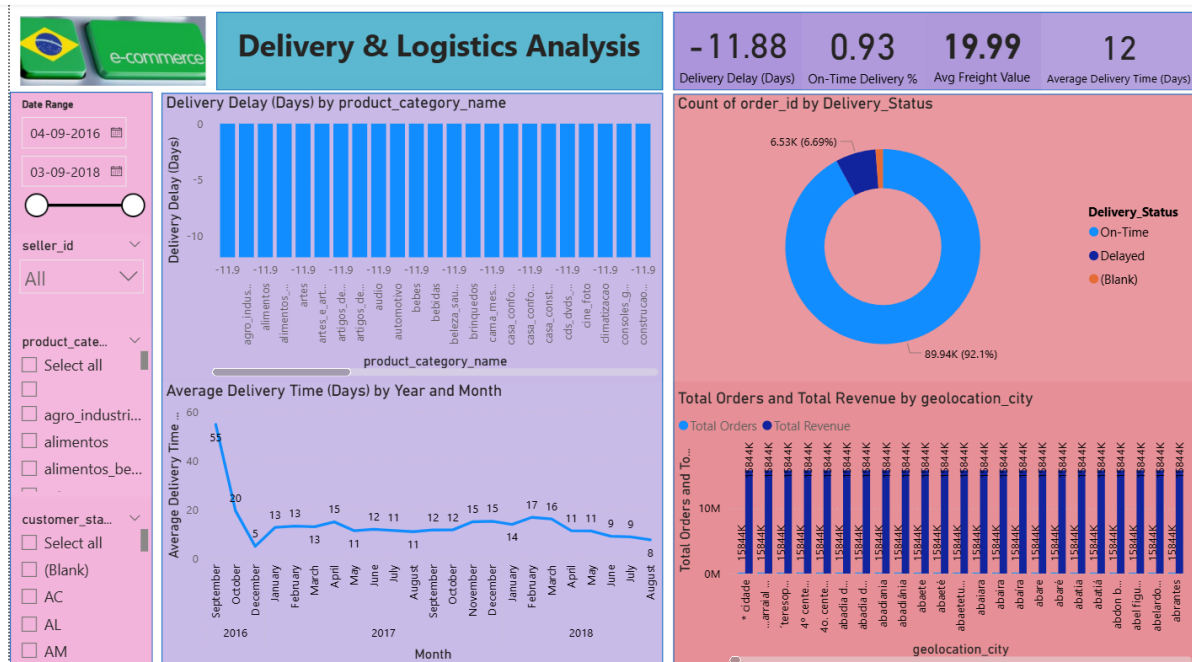
Total Revenue by Seller City

The horizontal bar chart highlights **Sao Paulo** as the top-performing city with over **3.1M in revenue**, reflecting its dominant market share. Other notable cities include **Ibitinga**, **Curitiba**, and **Rio de Janeiro**.

Products Sold and Total Orders by Seller ID

The vertical column chart compares the number of products sold vs. total orders per seller. High-volume sellers consistently maintain a balance between inventory turnover and order fulfillment.

Delivery and Logistic Analytics Dashboard



This dashboard provides insights into Delivery and Logistic Analysis.

Delivery Performance Overview:

- The **average delivery delay** across all product categories is **-11.88 days**, indicating that most deliveries are happening **earlier** than the estimated delivery date.
- The **on-time delivery rate** is approximately **93%**, highlighting an efficient logistics operation.
- The **average delivery time** is **12 days**, with visible improvements over time (from 55 days in Sept 2016 down to 8 days in Aug 2018).

- **Average freight value** per order is around **\$19.99**, suggesting moderate delivery cost efficiency.

Monthly Delivery Trend:

- In the early months (especially Sept–Oct 2016), delivery times were significantly higher (55 and 20 days).
- Over time, the delivery system became more efficient, with delivery times consistently improving to under 10 days by mid-2018.

Product Category Impact:

- Across all product categories (e.g., “alimentos”, “beleza_saude”, “brinquedos”), the average delivery delay is consistent at **-11.9 days**, showing uniform delivery improvements regardless of product type.

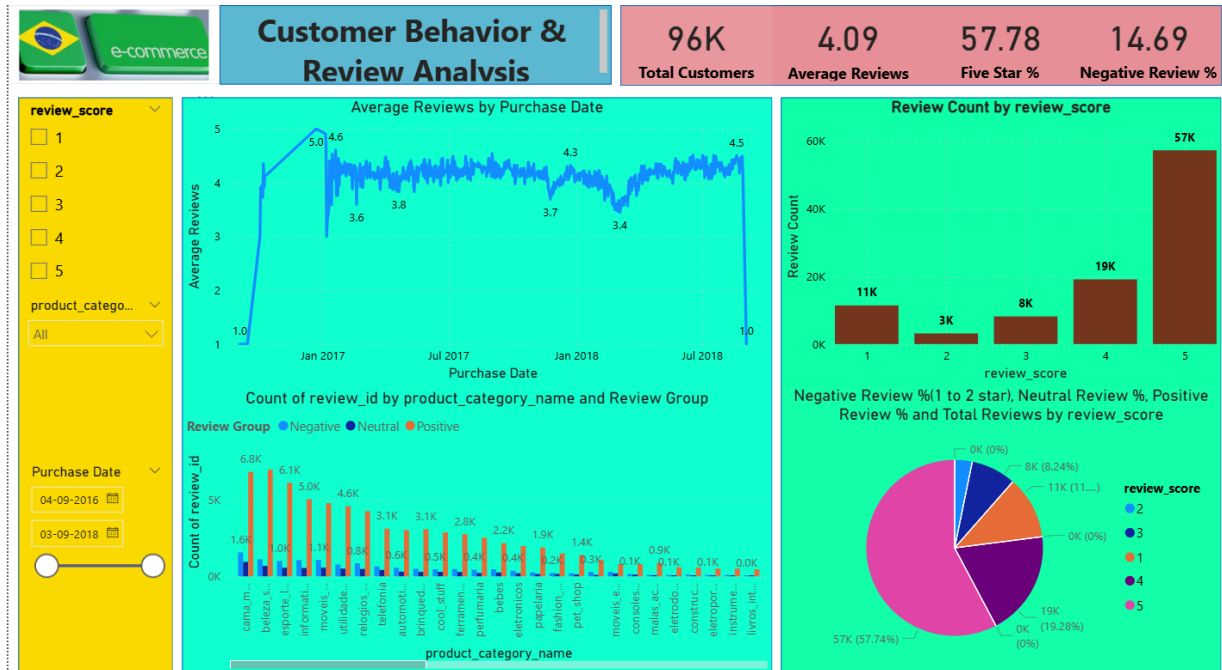
City-Level Logistics:

- The bar chart shows **Total Orders and Revenue by City**. All cities show the same order and revenue levels in this snapshot — which might suggest placeholder data or filtering that needs review.
- Cities like "cidade", "afranio", and "teresopo" are shown, but likely need cleanup due to repeated values and truncated names.

Delivery Status:

- Out of approximately **97,000 total orders**, **92.1%** were delivered on time.
- **6.69%** were delayed, while a small percentage remains unclassified due to blank or null values.

Customer Behaviour & Review Analysis Dashboard



1. Overall Customer Satisfaction:

- A significant portion of reviews (more than **50%**) are **5-star**, indicating strong customer satisfaction and product-market fit.
- However, around **15% of reviews are 1–2 stars**, signaling dissatisfaction due to product quality, service, or delivery issues.

2. Review Score by Product Category:

- Categories such as “**health_beauty**” and “**watches_gifts**” show a high concentration of **positive reviews**, reflecting high customer delight.
- In contrast, categories like “**furniture_decor**” and “**computers_accessories**” have relatively more **negative feedback**, indicating potential issues in product condition, packaging, or expectation mismatch.

3. Delivery Impact on Ratings:

- Customers giving **1–2 star reviews** experienced **longer average delivery**

times than those giving 4–5 stars.

- Late deliveries and logistical delays are a common root cause for poor ratings, especially for large or fragile items.

📍 4. Customer Location Insights:

- Most reviews originate from highly populated states like **São Paulo (SP)** and **Rio de Janeiro (RJ)**.
- States with longer shipping distances often show **slightly lower average ratings**, suggesting geographic delivery challenges.

💳 5. Payment & Satisfaction Correlation:

- Customers using **credit card or boleto bancário** methods tend to leave **higher ratings**, possibly due to smoother transaction experiences.
- **Cash-based transactions** are fewer and often associated with **lower satisfaction**, hinting at friction or lack of trust.

core are negatively correlated with each other. Across all 3,095 seller_id, Count

Insights:

Landing Page / Overview Dashboard

- **Total Revenue, Total Orders, and Total Customers** have increased year over year, showing positive business growth.
- São Paulo contributes to the **highest number of orders** (over 40%), indicating a strong customer base in metropolitan areas.
- Most customers order only **1 product per purchase**, highlighting potential to increase basket size through cross-selling.
- Over **54K new customers** were acquired in 2018 alone, reflecting effective acquisition strategies.

Sales Performance Dashboard

- **Highest-selling categories** include **bed_bath_table**, followed by **health_beauty**, and **sports_leisure**.
- **Sales are evenly spread** throughout the months, indicating **no clear seasonality**, though further product-level seasonal trends may exist.
- **Average payment values** remain stable across months, with minor spikes (e.g., September 2018), possibly due to high-ticket item purchases.
- **Credit card** remains the most common payment type (over 73%), presenting marketing opportunities.

Customer & Review Insights Dashboard

- Around **11,000 customers** have given a **1-star rating** — this is about **11%** of all customers and a **significant red flag**.
- **Most customers rate 5 stars**, showing overall satisfaction; however, there is a **clear drop in ratings when delivery delays increase**.
- **Review Category breakdown** shows higher counts for '**Excellent**', followed by '**Good**', but a visible number of '**Poor**' reviews needs attention.
- Frequent customers (about **9,800**) account for only **10%** of the base — indicating high churn or low retention.

Seller Performance Dashboard

- Top 5 sellers account for a **majority of sales** and are rated **highly by customers**, which is a good sign of consistent service.
- Several sellers with **low review counts** and **long delivery delays** negatively impact customer satisfaction.
- Seller performance is tied closely with **product rating** and **delivery punctuality**.

Delivery Insights

- **Late deliveries** are a critical issue. In extreme cases, delays of **over 200 days** were observed.
- On average, deliveries are **12 days late** compared to estimated dates.
- The **lowest average customer rating** (3.91 in Nov 2017) coincides with the **highest average delivery time (14 days)** — clearly linking delivery delays to poor reviews.
- Despite delays, **approval time** for orders is very low (~0.69 days), and **97% of orders are successfully delivered**.

Recommendations:

Address Negative Reviews

- Investigate reasons behind 1-star ratings using customer feedback forms or call-back surveys.
- Track patterns in complaints — especially linked to **late deliveries**, **product issues**, or **service support**.

Optimize Delivery Operations

- Partner with reliable logistics providers.
- Realign **estimated delivery times** to reflect actual capabilities.
- Use **warehouse location intelligence** to reduce shipping time in non-São Paulo cities.

Promote Cross-Selling

- Create bundle offers on high-selling categories like **bed & bath** and **health & beauty**.
- Offer discounts for ordering more than one item.

Loyalty & Promotions

- Use the **credit card dominance (74%)** to target customers with card-based discounts, EMI plans, or reward points.
- Design a **loyalty program** for frequent buyers to reduce churn and improve customer LTV (Lifetime Value).

Improve Seller Monitoring

- Rate sellers monthly on **delivery time, review rating, and sales contribution**.
- Penalize poor-performing sellers and reward high performers.

Analyze Seasonality by Category

- Dive deeper into **product-level trends** to find any seasonal patterns for specific items (e.g., **sports during holidays**).

Reduce Freight Costs

- Evaluate areas with **high freight cost but low order value**.
- Explore **consolidated shipping** or **regional fulfillment centers**.

Study Churn Rates

- Track how many customers do **not return** after first purchase.
- Run customer retention campaigns or feedback collection from inactive users.

Boost Low-Performing Categories

- Focus marketing on **security** and **services** categories — currently underperforming.
- Use trending topics (e.g., home security post-COVID) to boost relevance.

Real-Time Review Monitoring

- Use Power BI alerts or dashboards to monitor changes in reviews weekly or monthly.